Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/26/2008
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Readiness to Perform in a Space Analog Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dinges@pennmedicine.upenn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-9949
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Psychiatry		
PI Address 2:	423 Service Dr., 1013 Blockley Hall		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2005	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF00802		
Performance Goal No.:			
Performance Goal Text:			
	NASA Extreme Environment Mission Operations (NEEMO) has astronauts live in the NOAA Aquarius habitat (19 meters underwater off the coast of Florida). Aquarius is an excellent space analog environment for a number of challenges in the area of behavior and performance: confinement, limited personal space, isolation, workload timeline demands, EVAs, environmental risk, no quick return to surface in an emergency. Aquarius is operated by the National Underwater Research Center of the UNC at Wilmington. NEEMO missions work best for research that is directed toward proof of concept and feasibility studies in isolated and extreme environments. Research that needs testing in an analog mission works well in a NEEMO setting as it gives the researcher an idea of how the product works in a confined, but crowded environment focused on multiple mission accomplishments. The 18-day NEEMO 9 mission—from April 3 to April 20, 2006—was the longest NEEMO mission thus far. It was co-sponsored by NASA, the National Space Biomedical Research Institute (NSBRI), the Army's Telemedicine &		

Task Description:	Advanced Technology Research Center (TATRC), Canada's Centre for Minimal Access Surgery (CMAS) at the University of Ontario, Hamilton, as well as by and NOAA Undersea Research Center (NURC). Payload projects included: 1) Telesurgery and medical task performance experiments 2) Underwater mapping 3) EVA trials to test center of gravity configurations for the future exploration spacesuit 4) Feasibility study of behavioral performance measures This report contains the results of item 4, which included five substudies of the technical feasibility to assess the functioning of crewmembers (as individuals and as a team) in the close confinements and high tempo environment of NEEMO 9. The results were highly informative, as a number of the behavior/performance technologies worked very well in NEEMO 9. However, as in the case of all feasibility studies, the high-fidelity environment, including the configuration of Aquarius, the nature of the undersea environment, and the operational demands of the mission resulted in some hardware failures, timeline pressures, behavior monitoring difficulties, and consequently, some data loss. Thus, NEEMO 9 provided an excellent first-experience learning opportunity for the NASA behavior and performance area. The crew was outstanding in the time and effort they put into every performance assessment, and their willingness to adjust to the idiosyncratic aspects of some of the assessments. Habitat support staff and the topside team also contributed greatly to the success of the feasibility assessments. The NEEMO 9 performance technology feasibility study proved to be exceptionally valuable for identifying what works well in the behavioral area, what does not, and what could be improved to minimize astronaut burden and data loss. Data acquisition was also completed in NEEMO 12 (Vigilance, Stress and Sleep/Wake Measures in NEEMO 12 - A Simulated Space Environment). The goal of this study was to monitor performance, stress and sleep-wake patterns in 4 crew memebers and 3 Aquarius habitat personnel during the NEEMO 12 mission (March 6-17, 2007). This was accomplished through three specific aims; (1) test the usefulness of the 3-minute Psychomotor Vigilance Test; (2) acquire salivary cortisol measures of stress; and (3) monitor sleep-wake patterns using ambulatory technology (wrist actigraphs and sleep logs). Data acquisition in NEEMO 12 was successful. A total of 160 PVT performance tests were acquired, 153 salivary cortisol measures were obtained, and 40 days of wrist actigraphy, sleep-wake diary and subjective fatigue and stress scales were acquired, for a 98% adherence rate. The salivary cortisol and subjective ratings revealed no elevated stress during NEEMO 12, and the wrist actigraphy, sleep-wake diary, and subjective fatigue data from NEEMO 12 revealed no sleep los or elevated fatigue. These outcomes made the 3-minute PVT performance data from NEEMO 12 useful for adding to the normative PVT performance database for astronauts begun with PVT data from NEEMO 9. The resulting larger 3-minute PVT performance database on astronauts and aquanauts will be used to generate a PVT SelfTest user interface for use by astronauts in space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The neurobehavioral, psychosocial and physiological monitoring technologies used in NEEMO 9 and NEEMO 12 have considerable potential for many Earth-based applications, especially those situations in which monitoring of cognitive readiness, sleep need, stress levels, and team cohesion are needed to help ensure human effectiveness (e.g., in first responders, military personnel, and emergency personnel).
Task Progress:	NASA Extreme Environment Mission Operations (NEEMO) has astronauts live in the NOAA Aquarius habitat (19 meters underwater off the coast of Florida). Aquarius is an excellent space analog environment for a number of challenges in the area of behavior and performance: confinement, limited personal space, isolation, workload timeline demands, EVAs, environmental risk, no quick return to surface in an emergency. Aquarius is operated by the National Underwater Research Center of the UNC at Wilmington. NEEMO missions work best for research that is directed toward proof of concept and feasibility studies in isolated and extreme environments. Research that needs testing in an analog mission works well in a NEEMO setting as it gives the researcher an idea of how the product works in a confined, but crowded environment focused on multiple mission accomplishments. The 18-day NEEMO 9 mission—from April 3 to April 20, 2006—was the longest NEEMO mission thus far. It was co-sponsored by NASA, the National Space Biomedical Research Institute (NSBRI), the Army's Telemedicine & Advanced Technology Research Center (TATRC), Canada's Centre for Minimal Access Surgery (CMAS) at the University of Ontario, Hamilton, as well as by and NOAA Undersea Research Center (NURC). Payload projects included: (1) Telesurgery and medical task performance experiments ; (2) Underwater mapping ; (3) EVA trials to test center of gravity configurations for the future exploration spacesuit ; (4) Feasibility study of behavioral performance measures. This report contains the results of item 4, which included five substudies of the technical feasibility to assess the functioning of crewmembers (as individuals and as a team) in the close confinements and high tempo environment of NEEMO 9. The results were highly informative, as a number of the behavior/performance technologies worked very well in NEEMO 9. However, as in the case of all feasibility studies, the high-fidelity environment, including the configuration of Aquarius, the nature of the undersea environment, and the operational demands of the mission resulted in some hardware failures, timeline pressures, behavior monitoring difficulties, and consequently, some data loss. Thus, NEEMO 9 provided an excellent first-experience learning opportunity for the NASA behavior and performance area. The crew was outstanding in the time and effort they put into every performance assessment, and their willingness to adjust to the idiosyncratic aspects of some of the assessments. Habitat support staff and the topside team also contributed to the success of the feasibility assessments. The NEEMO 9 performance technology feasibility study proved to be exceptionally valuable for identifying what works well in the behavioral area, what does not, and what could be improved to minimize astronaut burden and data loss. Data acquisition was also completed in NEEMO 12 on 4 crew members and 3 Aquarius habitat personnel: 160 PVT performance tests were acquired, 153 salivary cortisol measures were obtained, and 40 days of wrist actigraphy, sleep-wake diary and subjective fatigue and stress scales were acquired, for a 98% adherence rate.

Bibliography Type:	Description: (Last Updated: 06/30/2026)
Awards	Dinges DF. "Laurence R. Young Space Biomedical Research Award for contributions to human performance in space, February 2008." Feb-2008
Awards	Dinges DF. "NASA Distinguished Public Service Medal, May 2007." May-2007